SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts Student List

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	A.Paheershana
Student ID	JF/ICT/24/04
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System required a facility to track and display the most recent successful login time of student accounts. This information helps administrators identify student account activity and understand when students last accessed the system.

Before this change, the system stored student account details and authentication information but did not record or display the last successful login time. Therefore, administrators could not easily identify whether a student had logged into the system recently.

The change was required to record the last successful login time for student accounts and display that information in the administrator's student list without affecting existing authentication or student management functionality.

1.2 Existing Behaviour

Before implementation:

- Student account information was stored in the database.
- Students could log in using their registered credentials.
- The authentication system verified usernames and passwords.
- The administrator could view student details through the Admin Dashboard.
- The system did not record the last successful login time for student accounts.
- The Admin Dashboard did not contain a Last Login column.
- Administrators could not distinguish between students who had logged in recently and students who had never logged in.

1.3 Requested Change

Before implementation:

- Student account information was stored in the database.
- Students could log in using their registered credentials.
- The authentication system verified usernames and passwords.
- The administrator could view student details through the Admin Dashboard.
- The system did not record the last successful login time for student accounts.
- The Admin Dashboard did not contain a Last Login column.
- Administrators could not distinguish between students who had logged in recently and students who had never logged in.

2. Implementation

2.1 Implementation Summary

CR-033 was implemented by extending the existing Student Management System backend and frontend.

A new `last_login` column was added to the `users` database table to store the most recent successful login date and time. The column allows a `NULL` value for accounts that have not yet recorded a successful login.

The user model was updated with an `updateLastLogin()` function. This function updates the timestamp for the corresponding user account after successful student authentication.

The authentication controller was modified to call the new function after the submitted username and password have been verified. Failed authentication attempts do not trigger the timestamp update.

The student model was updated to retrieve the `last_login` field by joining the `students` and `users` tables. This allows the existing student information endpoints to return the last login timestamp alongside the student details.

Finally, the Admin Dashboard was updated to display a Last Login column. A formatting function converts valid timestamps into a readable date and time. If the timestamp is missing or `NULL`, the dashboard displays “Never logged in”.

The implementation reuses the existing authentication and student retrieval functionality. No separate login-tracking API endpoint is required for this implementation.

2.2 Affected Files

File	Change	Purpose
database/schema.sql	Modified	Adds the last_login column to the users table for new database installations.
backend/src/models/userModel.js	Modified	Adds the updateLastLogin() function and exports it with the existing model functions.
backend/src/controllers/authController.js	Modified	Updates the last login timestamp after successful student authentication.
backend/src/models/studentModel.js	Modified	Retrieves the last_login field by joining student records with user accounts.
frontend/src/pages/DashboardPage.jsx	Modified	Adds the Last Login column and displays formatted login timestamps or “Never logged in”.

Database migration for existing installations: The existing `users` table must be updated once if the `last_login` column is not already present.

```
USE student_management;

ALTER TABLE users
ADD COLUMN last_login DATETIME NULL DEFAULT NULL;
```

This SQL statement should not be executed again if the column already exists.

2.3 Technical Changes

2.3.1 Database Modification

The users table was extended with a new field:

last_login DATETIME NULL DEFAULT NULL

The field stores the date and time of the most recent successful student login.

- DATETIME stores the login date and time.
- NULL indicates that no login timestamp has been recorded.
- DEFAULT NULL allows existing accounts and newly created accounts to remain without a recorded login time until a successful login occurs.

The change does not require a separate table because the login information belongs to an existing user account.

2.3.2 User Model Modification

The user model was updated to include the updateLastLogin() function.

The function executes a parameterized SQL statement to update the timestamp for a specific user ID.

UPDATE users

SET last_login = NOW()

WHERE user_id = ?;

The parameterized query ensures that the user ID is supplied separately from the SQL statement. The existing functions for finding users by username and ID are retained.

2.3.3 Authentication Controller Modification

The authentication controller was modified to record the last login time after the submitted credentials are verified successfully.

The controller checks the user's role and updates the timestamp for student accounts.

```
if (user.role === "STUDENT") {
   await updateLastLogin(user.user_id);
}
```

The existing JWT generation process and login response are retained.

If the username does not exist or the password is incorrect, the controller returns the existing authentication error response without updating the last login timestamp.

2.3.4 Student Model Modification

The student model was updated to retrieve the last login timestamp from the related user account.

The student information queries use an SQL join between the students and users tables.

```
FROM students AS s
INNER JOIN users AS u
   ON s.user_id = u.user_id
```

The relevant queries retrieve u.last_login along with the existing student fields.

This modification allows the dashboard to receive the login timestamp through the existing student retrieval functionality. Existing create, update, and delete operations are retained.

2.3.5 Admin Dashboard Modification

The Admin Dashboard was updated to include a Last Login column in the student list.

The frontend formatting function converts valid timestamps into a readable local date and time. When a student account has no recorded timestamp, the dashboard displays:

```
Never logged in
```

The new column is added without removing the existing Registration No., Name, Email, Phone, Edit, or Delete functionality.

2.3.6 Data Handling and Error Management

The implementation handles student accounts with a NULL login timestamp by displaying "Never logged in".

The dashboard also checks whether the timestamp can be parsed into a valid date before formatting it.

The database update is restricted to the authenticated student's user ID and is performed only after successful password verification.

The existing authentication error handling remains in place. The implementation does not expose password hashes or change the existing role-based access controls.

2.4 Implementation Logic

The implementation follows these steps:

1. Add the last_login column to the users table.
2. Retain the existing username and password authentication process.
3. Retrieve the user account using the submitted username.
4. Verify the submitted password against the stored password hash.
5. If authentication fails, return the existing error response without updating the timestamp.
6. If a student authenticates successfully, update last_login using the current database date and time.
7. Continue with the existing JWT generation and successful login response.
8. Retrieve student information together with the associated user account's last_login field.
9. Return the student information through the existing student retrieval endpoint.
10. Display the Last Login column in the Admin Dashboard.
11. Format a valid timestamp into a readable date and time.
12. Display "Never logged in" when the timestamp is NULL.
13. Verify that existing student management and authentication functionality continues to work.

2.5 Error and Data Handling

The feature extends the existing database and application logic without replacing the current authentication mechanism.

The implementation follows these data-handling rules:

- Successful student login: update the student's last login timestamp.
- Failed student login: do not update the timestamp.
- Student account with no recorded login: display “Never logged in”.
- Existing student information: preserve the existing fields and records.
- Existing student CRUD operations: retain the existing create, read, update, and delete functionality.
- Administrator authentication: retain the existing authentication process without applying the student-specific timestamp update.
- Missing or invalid timestamp: avoid displaying an invalid formatted date.

The implementation requires a database schema update and backend code changes. Therefore, unlike a frontend-only enhancement, this change modifies the database structure and backend authentication and retrieval logic.

3. Testing & Verification

3.1 Testing Approach

The implementation was designed to be verified through manual testing of the database, backend authentication process, student information retrieval, and Admin Dashboard.

Testing focuses on confirming that successful student logins update the timestamp, failed logins do not update it, the timestamp is displayed correctly, and existing functionality remains operational.

Testing Types:

- Functional testing
- Database testing
- Authentication testing

- API response testing
- UI testing
- Data retrieval testing
- Error-handling testing
- Regression testing

Environment: Development

Browser: Chrome / Microsoft Edge

Backend: Node.js and Express.js

Database: MySQL

Test Roles: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Verify the last_login column in the database.	The users table contains the new nullable timestamp field.	The last_login column was verified in the database.	Pass
TC-02	Log in successfully using a student account.	The student's last_login value is updated.	The timestamp was updated after successful student login.	Pass
TC-03	Log in using an incorrect password.	Authentication fails and the last login timestamp remains unchanged.	Login was rejected, and the timestamp remained unchanged.	Pass
TC-04	Attempt to log in using a nonexistent username.	Authentication fails without updating any student login timestamp.	Login was rejected without updating a student login timestamp.	Pass

TC-05	Log in successfully a second time.	The timestamp is updated to the latest successful login time.	The timestamp was updated to the latest successful login time.	Pass
TC-06	Retrieve the administrator's student list.	Student information includes the last_login field.	The student list included the last_login field.	Pass
TC-07	Open the Admin Dashboard.	The Last Login column appears in the student list.	The Last Login column was displayed correctly.	Pass
TC-08	View a student account with a NULL timestamp.	The dashboard displays “Never logged in”.	“Never logged in” was displayed correctly.	Pass
TC-09	View a student account with a valid timestamp.	The dashboard displays a readable date and time.	The login timestamp was displayed in a readable format.	Pass
TC-10	Navigate and search the student list.	Existing student listing and search functionality remain operational.	Student listing, navigation, and search worked correctly.	Pass
TC-11	Perform existing student Edit and Delete operations.	Existing student management functionality continues to work.	Student Edit and Delete operations continued to work correctly.	Pass
TC-12	Verify the existing login response and token generation.	Successful authentication continues to return the expected response and token.	Login returned the expected successful response and token.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	The database stores the last successful login timestamp for student accounts.	Database query	Met
AC-002	A successful student login updates the timestamp.	Login and database verification	Met
AC-003	A failed login does not update the timestamp.	Failed login and database comparison	Met
AC-004	Student information retrieval includes the last login field.	API response inspection	Met
AC-005	The Admin Dashboard displays a Last Login column.	UI inspection	Met
AC-006	Accounts without a recorded login display “Never logged in”.	UI and database verification	Met
AC-007	The displayed timestamp is readable and correctly formatted.	UI inspection	Met
AC-008	Existing authentication and token generation remain operational.	Regression testing	Met
AC-009	Existing student CRUD functionality remains operational.	CRUD regression testing	Met

5. Evidence

Evidence 01 – Database Last Login Column

Related Acceptance Criteria: AC-001

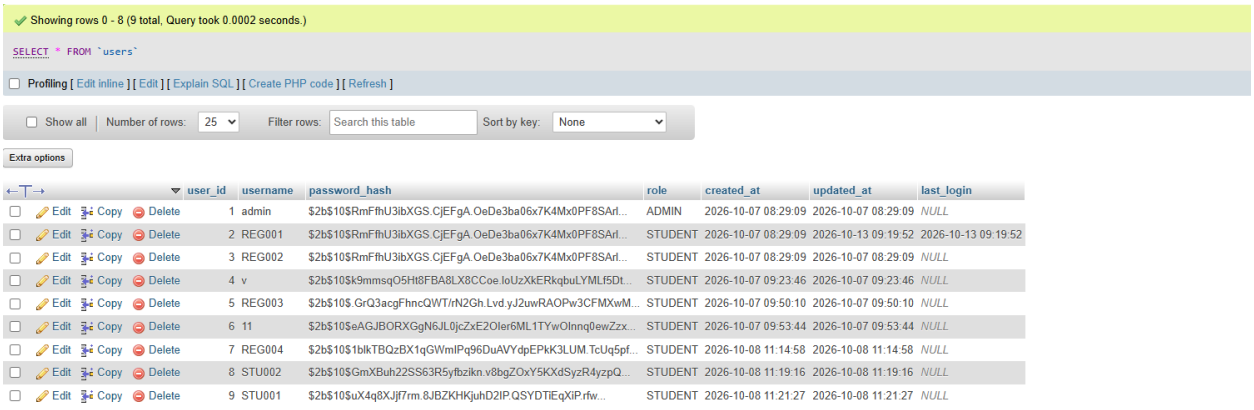
Description: Insert a screenshot from MySQL Workbench or the database client showing the last_login column in the users table.

Suggested SQL:

USE student_management;

DESCRIBE users;

Figure 1: Last login field in the users database table.



Showing rows 0 - 8 (9 total, Query took 0.0002 seconds.)

```
SELECT * FROM `users`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	user_id	username	password_hash	role	created_at	updated_at	last_login
Edit Copy Delete	1	admin	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	ADMIN	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	2	REG001	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-13 09:19:52	2026-10-13 09:19:52
Edit Copy Delete	3	REG002	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	4	v	$2b$10$k9mmsqO5Ht8FBA8LX8CCoe.loUzXkERkqbuLYMLf5Dt...	STUDENT	2026-10-07 09:23:46	2026-10-07 09:23:46	NULL
Edit Copy Delete	5	REG003	$2b$10$.GrQ3acgFhncQWT/rN2Gh.Lvd.yJ2uwRAOPw3CFMXwM...	STUDENT	2026-10-07 09:50:10	2026-10-07 09:50:10	NULL
Edit Copy Delete	6	11	$2b$10$eAGJBORXGgN6JL0jcZxE2Oler6ML1TYwOlnnq0ewZzx...	STUDENT	2026-10-07 09:53:44	2026-10-07 09:53:44	NULL
Edit Copy Delete	7	REG004	$2b$10$1blkTBQzBX1qGWmIPq96DuAVYdpEPkK3LUM.TcUq5pf...	STUDENT	2026-10-08 11:14:58	2026-10-08 11:14:58	NULL
Edit Copy Delete	8	STU002	$2b$10$GmXBuh22SS63R5yfbzikn.v8bgZOxY5KXdSyzR4yzpQ...	STUDENT	2026-10-08 11:19:16	2026-10-08 11:19:16	NULL
Edit Copy Delete	9	STU001	$2b$10$uX4q8XJjf7rm.8JBZKHKjuhD2IP.QSYDTiEqXiP.rfw...	STUDENT	2026-10-08 11:21:27	2026-10-08 11:21:27	NULL

Evidence 02 – Successful Student Login

Related Acceptance Criteria: AC-002, AC-008

Description: Insert screenshots showing a successful student login and the corresponding updated timestamp in the database.

Suggested SQL:

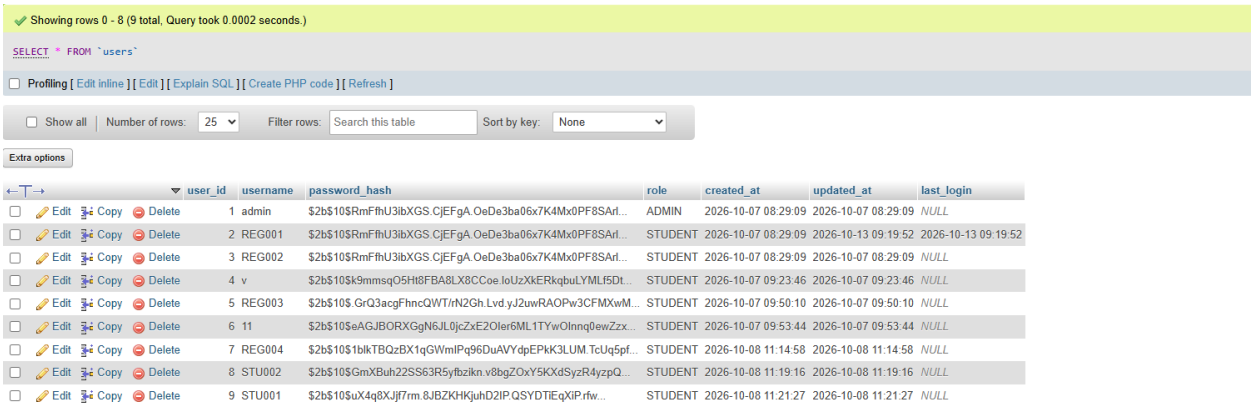
SELECT user_id, username, role, last_login

FROM users

WHERE role = 'STUDENT'

ORDER BY user_id;

Figure 2: Last login timestamp updated after successful student authentication.



Showing rows 0 - 8 (9 total, Query took 0.0002 seconds.)

```
SELECT * FROM `users`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

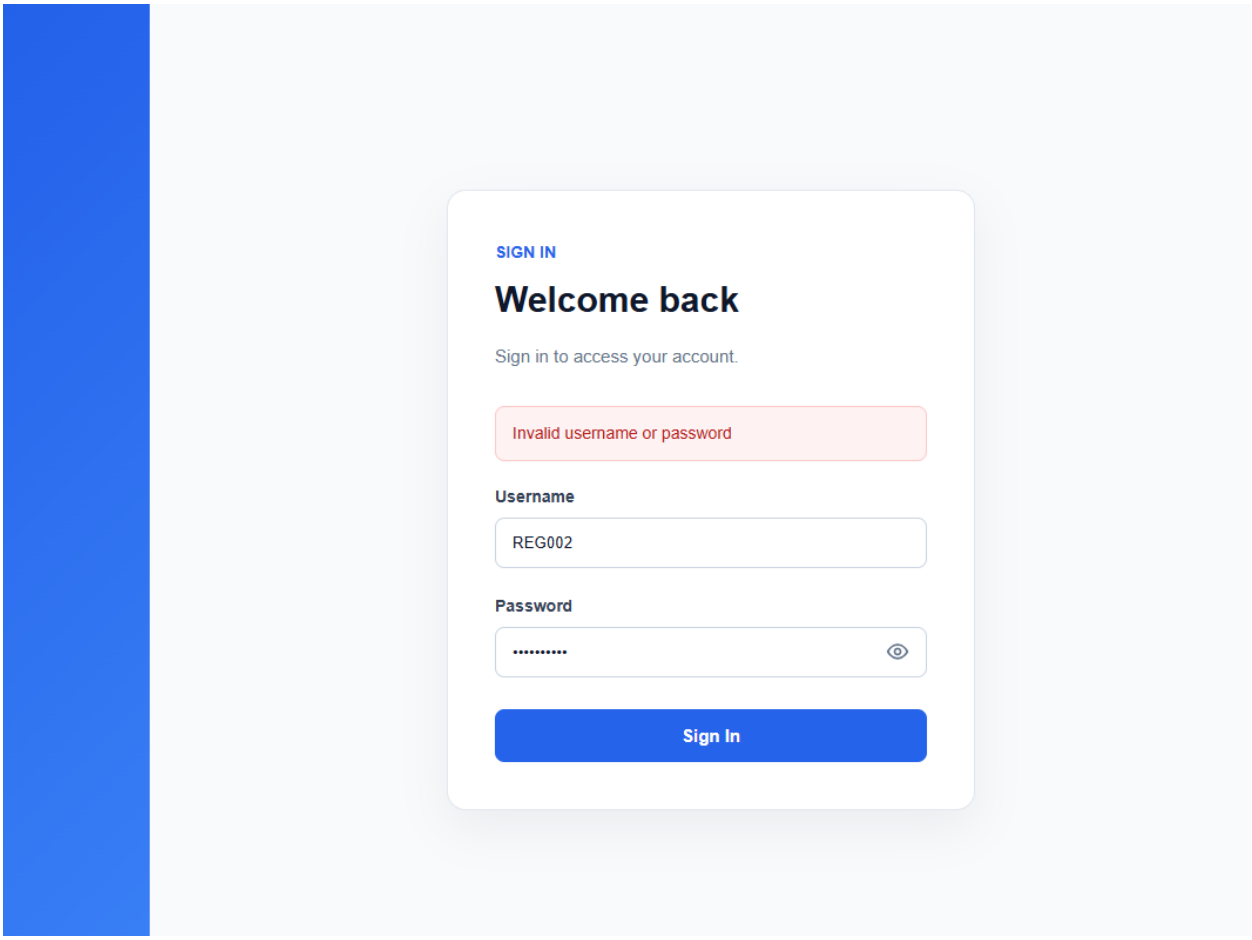
	user_id	username	password_hash	role	created_at	updated_at	last_login
Edit Copy Delete	1	admin	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	ADMIN	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	2	REG001	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-13 09:19:52	2026-10-13 09:19:52
Edit Copy Delete	3	REG002	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	4	v	$2b$10$k9mmsqO5Ht8FBA8LX8CCoe.loUzXkERkqbuLYMLf5Dt...	STUDENT	2026-10-07 09:23:46	2026-10-07 09:23:46	NULL
Edit Copy Delete	5	REG003	$2b$10$.GrQ3acgFhncQWT/rN2Gh.Lvd.yJ2uwRAOPw3CFMXwM...	STUDENT	2026-10-07 09:50:10	2026-10-07 09:50:10	NULL
Edit Copy Delete	6	11	$2b$10$eAGJBORXGgN6JL0jcZxE2Oler6ML1TYwOInnq0ewZzx...	STUDENT	2026-10-07 09:53:44	2026-10-07 09:53:44	NULL
Edit Copy Delete	7	REG004	$2b$10$1blkTBQzBX1qGWmIPq96DuAVYdpEPkK3LUM.TcUq5pf...	STUDENT	2026-10-08 11:14:58	2026-10-08 11:14:58	NULL
Edit Copy Delete	8	STU002	$2b$10$GmXBuh22SS63R5yfbzikn.v8bgZOxY5KXdSyzR4yzpQ...	STUDENT	2026-10-08 11:19:16	2026-10-08 11:19:16	NULL
Edit Copy Delete	9	STU001	$2b$10$uX4q8XJjf7rm.8JBZKHKjuhD2IP.QSYDTiEqXiP.rfw...	STUDENT	2026-10-08 11:21:27	2026-10-08 11:21:27	NULL

Evidence 03 – Failed Login Does Not Update Timestamp

Related Acceptance Criteria: AC-003

Description: Insert screenshots showing a failed login attempt and database evidence confirming that the relevant student's last login timestamp has not changed.

Figure 3: Last login timestamp remains unchanged after failed authentication.



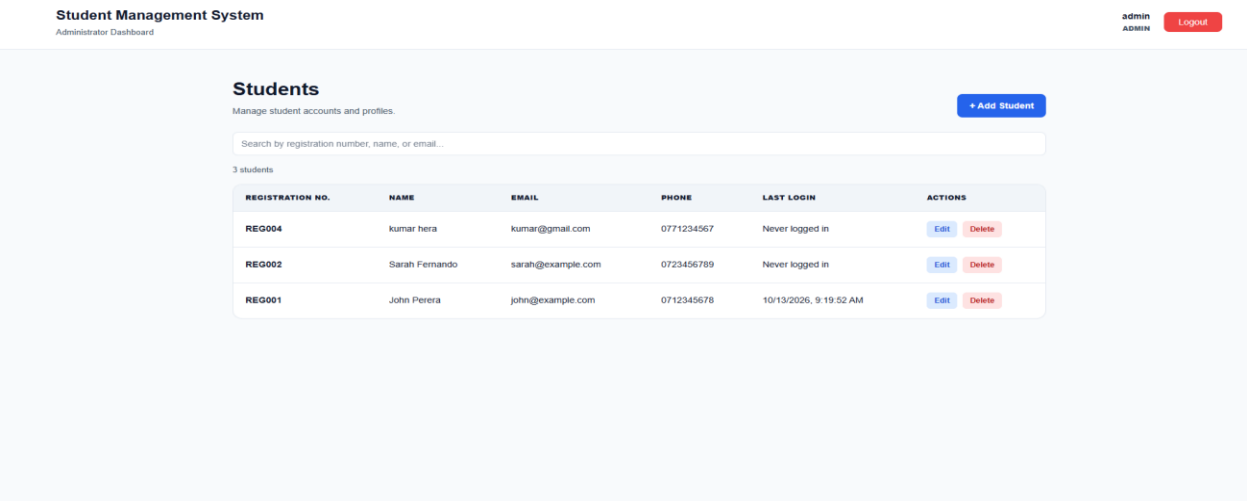
	user_id	username	password_hash	role	created_at	updated_at	last_login
Edit Copy Delete	1	admin	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	ADMIN	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	2	REG001	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-13 09:19:52	2026-10-13 09:19:52
Edit Copy Delete	3	REG002	$2b$10$RmFfhU3ibXGS.CjEFgA.OeDe3ba06x7K4Mx0PF8SArl...	STUDENT	2026-10-07 08:29:09	2026-10-07 08:29:09	NULL
Edit Copy Delete	4	v	$2b$10$k9mmsqO5Ht8FBA8LX8CCoe.IoUzXkERkqbuLYMLf5Dt...	STUDENT	2026-10-07 09:23:46	2026-10-07 09:23:46	NULL
Edit Copy Delete	5	REG003	$2b$10$.GrQ3acgFhncQWT/rN2Gh.Lvd.yJ2uwRAOPw3CFMXwM...	STUDENT	2026-10-07 09:50:10	2026-10-07 09:50:10	NULL
Edit Copy Delete	6	11	$2b$10$eAGJBORXGgN6JL0jcZxE2Oler6ML1TYwOlnnq0ewZzx...	STUDENT	2026-10-07 09:53:44	2026-10-07 09:53:44	NULL
Edit Copy Delete	7	REG004	$2b$10$1blkTBQzBX1qGWmlPq96DuAVYdpEPkK3LUM.TcUq5pf...	STUDENT	2026-10-08 11:14:58	2026-10-08 11:14:58	NULL
Edit Copy Delete	8	STU002	$2b$10$GmXBuh22SS63R5yfbzikn.v8bgZOxY5KXdSyzR4yzpQ...	STUDENT	2026-10-08 11:19:16	2026-10-08 11:19:16	NULL
Edit Copy Delete	9	STU001	$2b$10$uX4q8XJjf7rm.8JBZKHKjuhD2IP.QSYDTiEqXiP.rfw...	STUDENT	2026-10-08 11:21:27	2026-10-08 11:21:27	NULL

Evidence 04 – Last Login Column in Admin Dashboard

Related Acceptance Criteria: AC-004, AC-005, AC-007

Description: Insert a screenshot of the administrator's student list showing the new Last Login column and a readable date and time for a student who has logged in.

Figure 4: Administrator dashboard displaying student last login information.

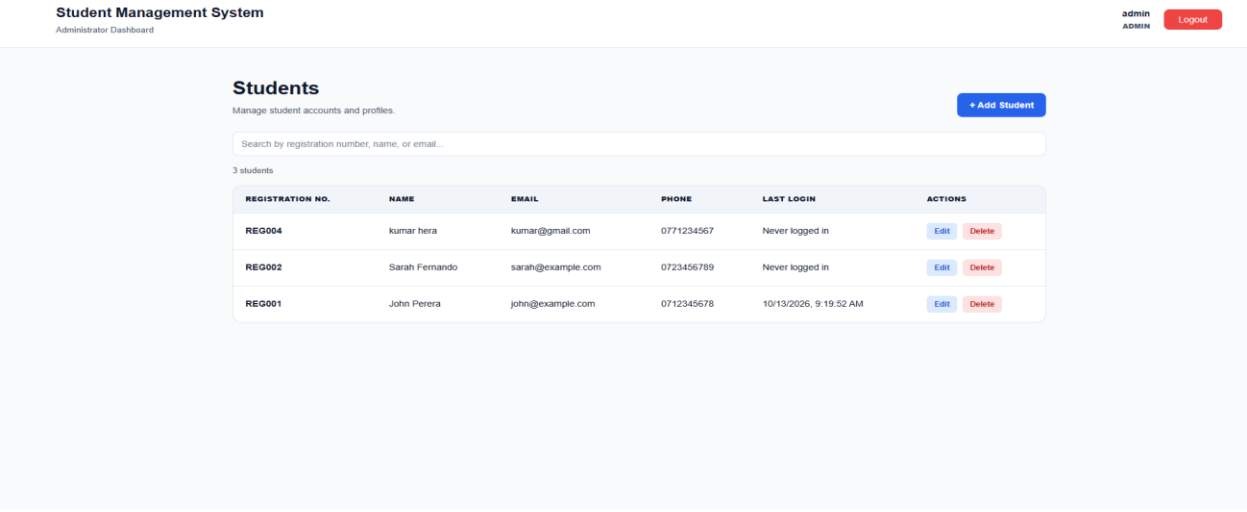


Evidence 05 – Never Logged In State

Related Acceptance Criteria: AC-006

Description: Insert a screenshot showing a student account with no recorded login timestamp. The dashboard should display “Never logged in”.

Figure 5: Student account with no recorded login history.



6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Theme Persistence	Verified
Existing Functionality	Verified
Final Status	Accepted

6.1 Final Summary

CR-033 introduces last login tracking and display functionality for student accounts in the Student Management System.

The implementation adds a last_login field to the users database table and extends the user model to update this field after successful student authentication. The authentication controller performs the update after password verification, while failed login attempts do not trigger a timestamp update.

The student model retrieves the timestamp through a join between the students and users tables. The Admin Dashboard displays the information in a dedicated Last Login column. A readable date and time are displayed for accounts with a recorded login, while accounts without a timestamp display “Never logged in”.

The change is designed to preserve existing login, token generation, student retrieval, search, Edit, and Delete functionality. It requires a database migration and modifications to the existing backend and frontend files.

The implementation is prepared. Final completion and acceptance should be recorded after the database migration, successful and failed login scenarios, dashboard display, and regression tests have been executed and their results documented.

6.2 Issues / Limitations

1. The last_login column must exist in the database before the updated backend code can use it.
2. Existing accounts will show “Never logged in” until a successful student login is recorded.
3. The feature records the most recent successful student login only; it does not maintain a full login history.
4. The displayed time uses the browser's date and time formatting. The database server's time zone should be considered when checking exact timestamp values.
5. The current implementation updates the timestamp for student accounts only. Administrator login tracking is not included in this change request.
6. The test results and final acceptance status must be based on actual execution in the development environment.

Student Name: _____A.Paheershana_____

Student ID: _____JF/ICT/24/04_____

Date: 13 October 2026

Signature: _____A.Paheershana_____

~ END OF REPORT ~